HipMRI 3D Improved U Net Feasibility Review Version 4

Extending a verified 2D implementation to a patient separated 3D comparison

Dataset	Difficulty	Target
HipMRI 3D NIfTI, 211 scans and 38 patients	**Hard**	**Patient mean prostate Dice at least 0.70**
Baseline	**Proposed**	**Split**
Plain 3D U Net, course Ref. 4	**Residual deep supervised 3D U Net, course Ref. 2**	**Seed 42, 27 train, 6 validation, 5 test patients**

User need and engineering decision

A radiotherapy contouring specialist needs a reviewable draft of prostate and organ at risk boundaries, not an autonomous treatment decision. The decision is whether volumetric context, residual units and deep supervision reduce base, apex, cross slice and adjacent organ errors enough to justify additional GPU memory and runtime. The completed 2D work supplies the engineering workflow; HipMRI performance is evaluated independently in 3D.

Scope and five acceptance criteria

1 Zero patient overlap between train, validation and test. 2 MRI and mask pairs pass shape, affine, finite value and integer label checks. 3 Held out patient mean prostate Dice is at least 0.70, with per class Dice and IoU reported. 4 Peak GPU VRAM, parameter count and per case inference time are recorded. 5 Three to five held out failures are reviewed with input, ground truth, prediction and prostate false positive and false negative overlays.

Model choice course concepts and source evidence

Dataset and label provenance The CSIRO collection **Labelled weekly MR images of the male pelvis** identifies six indexed mask classes. Its supporting BriefDataDescription.txt states **5 = Prostate** in the Labels block. This authoritative mapping, DOI 10.25919/45T8-P065, determines the target label; it is not inferred from appearance, region size or numbering order. **Architecture** Çiçek et al. support volumetric convolution and encoder decoder skips for the plain 3D baseline. Isensee et al. provide the course basis for residual feature refinement and training only deep supervision in the Hard model. Weighted cross entropy plus soft Dice addresses six class segmentation and the small prostate foreground.

Preliminary feasibility evidence

The data audit records 211 scans from 38 patients and a fixed seed 42 split of 27, 6 and 5 patients with zero overlap. Local synthetic NIfTI tests verify recursive pairing, invalid data rejection, patient separation, finite 3D forward and backward passes, auxiliary outputs and validation only checkpoint selection. Earlier OASIS 2D training verifies the reusable training and prediction workflow only; its scores are not HipMRI evidence.

Risks planned compute budget and fallback

Risks are 3D memory pressure, small foreground crops, orientation or alignment errors, slow I/O and failed cluster jobs. The planned budget is one short Rangpur GPU smoke run followed by two matched full runs, plain and improved, on one GPU at a time. Use batch size 1 and mixed precision only after the smoke check; select checkpoints on validation data and do not inspect test cases during model selection. If the improved model provides no reliable boundary benefit at acceptable cost, retain the plain 3D U Net.

Questions for tutor confirmation

1. Confirm that extending the completed 2D implementation into a matched plain-versus-improved 3D U-Net study satisfies the Hard scope.
2. Confirm the patient-level evaluation protocol and whether prostate Dice >= 0.70 is assessed with background reported separately.
3. Confirm the expected GPU budget, patch strategy and midpoint evidence required before full training.

References

[1] CSIRO, Labelled weekly MR images of the male pelvis, v2, DOI 10.25919/45T8-P065, BriefDataDescription.txt, Labels block. [2] Isensee et al., Brain Tumor Segmentation and Radiomics Survival Prediction, arXiv:1802.10508, 2018. [3] Dowling et al., Int J Radiat Oncol Biol Phys 93(5), 2015, DOI 10.1016/j.ijrobp.2015.08.045. [4] Çiçek et al., 3D U Net, MICCAI 2016, arXiv:1606.06650.

Appendix A 2D Implementation Evidence

Supporting evidence for the reusable 2D workflow and not HipMRI performance

The 2D stage verifies the segmentation training and evaluation workflow. The HipMRI stage reuses that engineering experience while introducing a new dataset, clinical target, 3D data pipeline and 3D models. Trained 2D weights and OASIS scores are not transferred as HipMRI evidence.

Evidence retained from the 2D implementation

Dataset	Training	Best validation DSC	Test foreground mean DSC
OASIS brain MRI slices	80 epochs on A100	0.9770 at epoch 29	0.9694

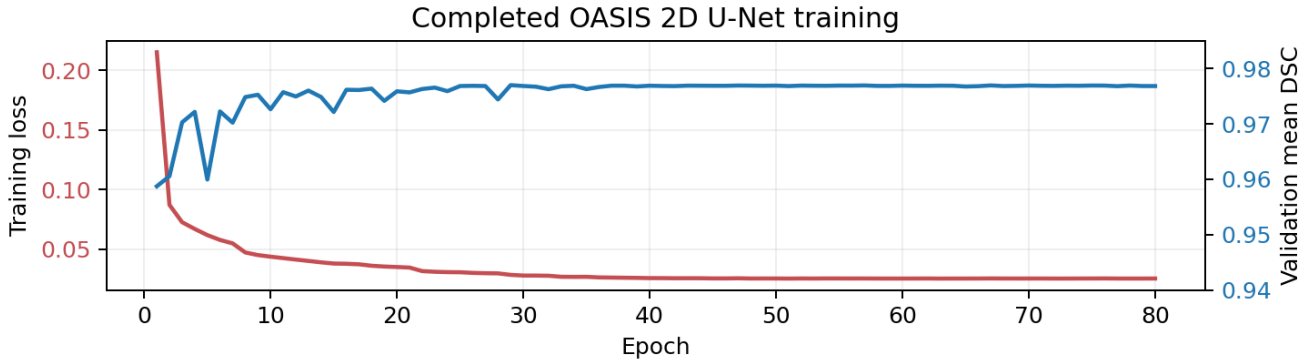


Figure 1. Training loss decreases while validation mean DSC stabilises near 0.977.

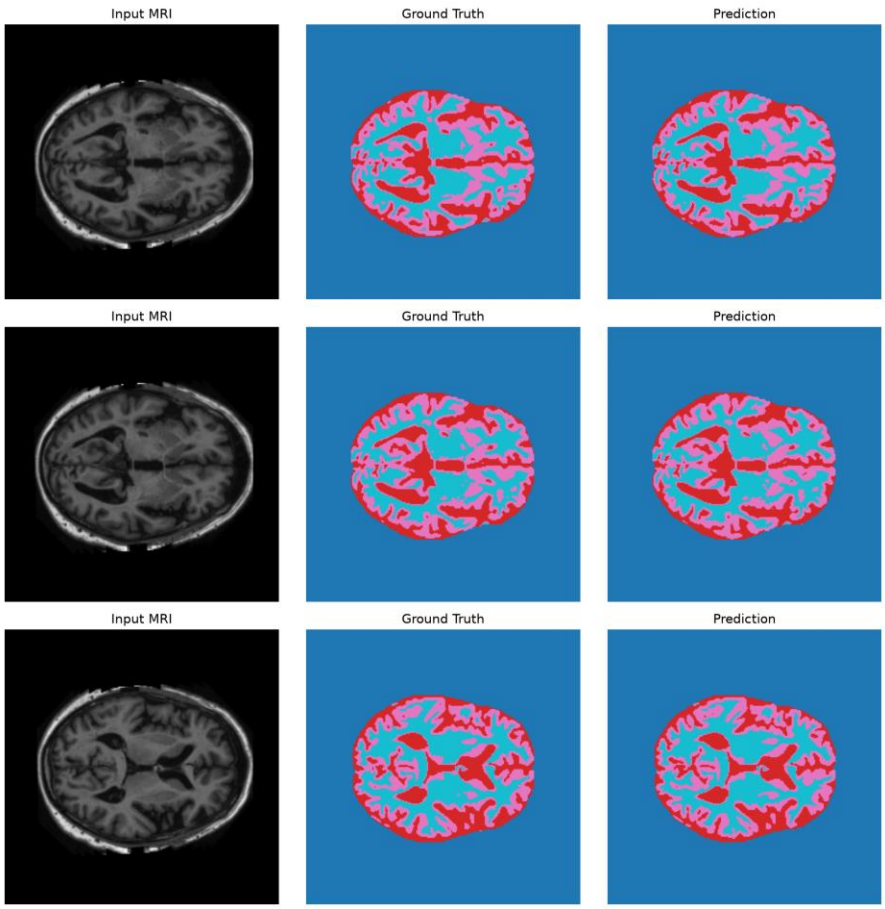


Figure 2. Held-out OASIS examples: input MRI, ground truth and 2D U-Net prediction.

Appendix B HipMRI 3D Implementation Plan

Supporting implementation status and matched 3D experiment plan

How the 2D implementation supports the 3D stage

The reusable part of the 2D work is the engineering workflow for training, validation, checkpoints and prediction review. HipMRI replaces slice-based loading and 2D operations with volumetric NIfTI loading, patient-level separation, 3D convolution and full-volume reconstruction. This extends the implementation; it does not convert trained 2D weights.

Stage	Status	Evidence
2D OASIS implementation	Complete	Training, checkpoint reload and held-out prediction review retained as engineering evidence.
3D HipMRI data pipeline	Implemented	Volumetric NIfTI pairing; shape, affine, finite-intensity and integer-label validation.
Patient split	Implemented	Seed-42 train/validation/test manifest with zero-overlap checks.
Matched 3D models	Implemented	Plain baseline and residual improved model with two training-only auxiliary heads.
Automated 3D verification	8 tests pass	Pairing, patient split, forward/backward, loss and evaluation logic.
Real HipMRI evaluation	Next at midpoint	Data audit, Rangpur GPU smoke run, then matched baseline and improved full runs.

Plain 3D baseline and improved 3D comparison

Reuse verified 2D workflow	Adapt HipMRI 3D data pipeline	Freeze patient manifest	Train matched 3D models	Held-out 3D metrics and failure panels

A plain 3D U-Net first establishes the HipMRI baseline. The Improved 3D U-Net then adds residual units and two training-only deep-supervision heads. Both models use the same patient split, preprocessing, optimisation settings and validation-based checkpoint selection.

Final evidence will report patient-level prostate Dice with a target of at least 0.70, per-class Dice/IoU, peak VRAM, per-case inference time and 3-5 held-out boundary-error cases. Checkpoint selection uses validation data only; the 2D results remain separate feasibility evidence.

Resource risk and fallback

A short smoke run will determine patch size, batch size and mixed precision before full Slurm jobs. If the improved model does not reduce real held-out errors at acceptable cost, the plain 3D U-Net remains the supported fallback.